Available online at www.sciencedirect.com

SCIENCE DIRECT®

Brain Research 1052 (2005) 71 – 81

BRAIN RESEARCH

www.elsevier.com/locate/brainres

Research Report

Comparative analysis of cortical layering and supragranular layer enlargement in rodent carnivore and primate species

Jeffrey J. Hutsler[a,b,*], Dong-Geun Lee[a], Kristin K. Porter[c]

[a]*Department of Psychology, 525 E. University Avenue, University of Michigan, Ann Arbor, MI 48109-1109, USA*
[b]*Program in Neuroscience, University of Michigan, Ann Arbor, MI, USA*
[c]*University of Alabama School of Medicine, Birmingham, AL, USA*

Accepted 5 June 2005
Available online 12 July 2005

Abstract

The mammalian cerebral cortex is composed of individual layers characterized by the cell types they contain and their afferent and efferent connections. The current study examined the raw, and size-normalized, laminar thicknesses in three cortical regions (somatosensory, motor, and premotor) of fourteen species from three orders of mammals: primates, carnivores, and rodents. The proportional size of the pyramidal cell layers (supra- and infragranular) varied between orders but was similar within orders despite wide variance in absolute cortical thickness. Further, supragranular layer thickness was largest in primates (46 ± 3 percent), followed by carnivores (36 ± 3 percent), and then rodents (19 ± 4 percent), suggesting a distinct difference in the proportion of cortex devoted to corticocortical connectivity across these orders. Although measures of supragranular layer thickness are highly correlated with measures of overall brain size, such associations are not present when independent contrasts are used to control for phylogenetic inertia. Interestingly, neurogenesis time span remains strongly associated with supragranular layer thickness despite size normalization and controlling for phylogenetic inertia. Such layering differences between orders, and similarities amongst species within an order, suggest that supragranular layer expansion may have occurred early in mammalian evolution and may be related to ontogenetic variables such as neurogenesis time span rather than measures of overall size.

Theme: Other systems of the CNS
Topic: Comparative neuroanatomy

Keywords: Column; Cortex; Mammal; Evolution

1. Introduction

Cortical neurons are organized into horizontal layers and vertical columns. Cortical columns range from the pial surface of the cerebral cortex and traverse the horizontal cortical layers perpendicularly down to the white matter [9,12,26,47,53]. Each of these vertical columns are a single cell in diameter, and these smaller columns aggregate together to form macrocolumns [7,35,51]. Although columns are thought to be the fundamental processing unit of cerebral cortex [26,52,53], both the precise function of columnar organization [26] and the degree to which genes control this organization [1] remain unclear.

The cortical uniformity hypothesis, which was initially based upon the number of cells within a column [72], argued for homology of columnar structure across a wide variety of mammalian species [51,63,64,72] (see Fuster [25] for a review). Differences between mammals were ascribed to changes in the total number of columnar units within the cortical sheet [64,75,78]. Subsequent studies of cortical organization and columnar structure have demonstrated that

* Corresponding author. Fax: +1 734 763 7480.
E-mail address: hutsler@umich.edu (J.J. Hutsler).

0006-8993/$ - see front matter © 2005 Elsevier B.V. All rights reserved.
doi:10.1016/j.brainres.2005.06.015

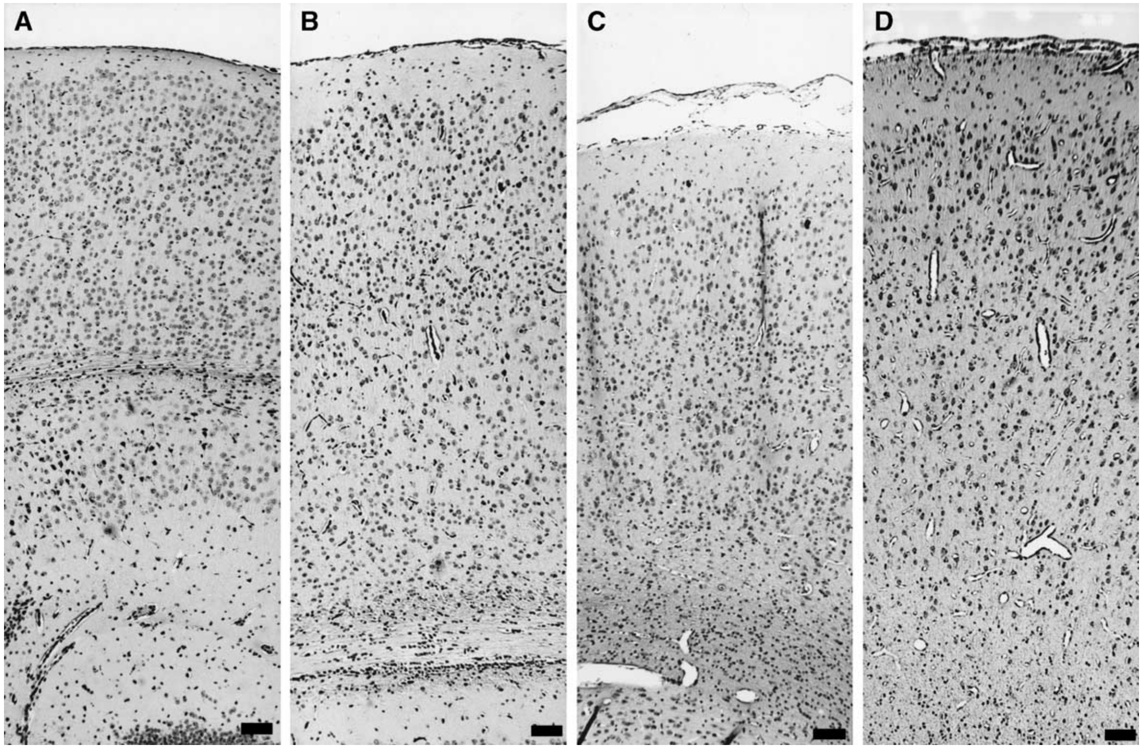

Fig. 1. Cross sections of the cerebral cortex in four species shown at the same magnification: (A) mouse; (B) rat; (C) ferret; and (D) cat. The absolute thickness of cortex can vary substantially within a single order (e.g. mouse vs. rat). Scale bar = 100 μm.

when a broad range of mammals is considered, columnar cell counts, neurochemical expression, and even connectional patterns can vary across species as well as across cortical locations within a species [4,5,20,47,57,58].

In addition to columnar organization, the mammalian cortex consists of several distinct lamina that receive and send projections to specific sets of targets [49]. The deepest cortical layers-the infragranular layers (V and VI)-mature early and project primarily to subcortical targets, while the most superficial layers-the supragranular layers (II and III)-mature last [49] and project primarily to other cortical locations [13,25,34]. This organization has led several authors to suggest that the supragranular layers and their associated cortical projections participate heavily in higher cognitive functions by forming networks of primary, secondary, and association areas that act together to generate a variety of behaviors [51,68].

In the present study, we further assess the variability of cortical organization across species by examining the absolute and size-normalized laminar thicknesses of fourteen mammalian species, taken from three orders: primates, carnivores, and rodents. Variance in the laminar organization of cortical columns across species may indicate differences in the internal structure of columns, or differences in the pattern of afferent and efferent connections that embed a column within a larger network of cortical and subcortical regions. Such variance would also suggest that the intrinsic structure of cortical columns may have diverged during mammalian evolution, resulting in unique patterns of intrinsic organization across species [47] and across locations within a single species [32,78].

2. Materials and methods

2.1. Tissue collection and histological processing

Tissue specimens were acquired from fourteen adult species representing three mammalian orders. These included five primate species (human, gorilla, chimpanzee, squirrel monkey, and rhesus macaque), three carnivore species (dog, cat, and ferret), and six rodent species (rat, mouse, guinea pig, woodchuck, porcupine, and capybara). Because these specimens often came from exotic mammals, multiple sources were utilized (see Table 1). Regions from three cortical locations were examined: primary somatosensory cortex (koniocortex); primary motor cortex (agranular cortex); and premotor cortex (dysgranular cortex). The location of each region was first determined using published regional maps and stereotaxic atlases [2,6,10,18,19,22,41,42,45,56,77,79] and was subsequently confirmed by using the architectonic characteristics of each cortical type [3]. Since we did not have access to the whole brain in several of the more exotic species, only primary somatosensory cortex was available in every species examined.

Although cell counts can vary depending upon the specific histological method employed, cortical layering can be reliably discerned by a variety of procedures. In the

Table 1
Species and sources

Order	Common Name	Scientific Name	*n*	Sources (Quantity)
Primate	Human	Homo sapiens	3	Dartmouth Hitchcock Medical Center
	Gorilla	Gorilla gorilla	1	Harvard Univ.
	Chimpanzee	Pan troglodytes	3	Harvard Univ. (1); Yakovlev Collection (2)
	Rhesus Macaque	Macaca mulatta	4	UC Davis (1); Yakovlev Collection (3)
	Squirrel Monkey	Saimiri sciureus	2	Yakovlev Collection
Carnivore	Dog	Canis (lupus) familiaris	3	Harvard Univ.
	Ferret	Mustela furo	1	Smith College
	Cat	Felis silvestris catus	2	UC Davis (1); Yakovlev Collection (1)
Rodent	Woodchuck	Marmota monax	1	Harvard Univ.
	Porcupine	Erethizon dorsatum	1	Harvard Univ.
	Capybara	Hydrochoerus hydrochaeris	3	Yakovlev Collection
	Rat	Rattus rattus norwegicus	3	Pfizer, Inc.
	Guinea Pig	Cavia porcellus	2	Harvard Univ. (1); Yakovlev Collection (1)
	Mouse	Mus musculus	3	Pfizer, Inc.

present study, tissue processing varied by the source from which it was acquired. Specimens acquired from the Yakovlev collection and local laboratories were processed with standard thionin or cresyl violet-based Nissl procedures [39], while human specimens were processed with a hemotoxylin and eosin stain commonly used for postmortem human specimens (see below). To compensate for variance in postmortem material, especially when different methods and processing locations are utilized, all individual layer measures were size normalized to the total cortical thickness of each specimen.

Mammalian animal specimens were postfixed for a minimum of 48 hours in 4% paraformaldehyde. Prior to sectioning on a freezing microtome, blocks were cryoprotected in phosphate buffer with 25–30% sucrose. Eighty sections from each block were cut at 40 μm on a sliding microtome. Cut sections were mounted from gel alcohol onto gelatin-coated slides and allowed to air dry for at least 24 hours. Every third tissue section was rehydrated through a series of graded alcohols to water and stained with 0.05% thionin. The sections were then differentiated in alcohol and, if needed, returned to the thionin for additional staining. Following dehydration and clearing, sections were coverslipped with Permount.

For the hemotoxylin and eosin procedure, tissue blocks were postfixed in 4% paraformaldehyde and then embedded in paraffin prior to being sectioned at 8 μm on a rotary microtome. These sections were then deparaffinized, hydrated to water, and processed in hematoxylin for fifteen minutes, followed by a wash in running tap water for twenty minutes prior to being counter stained with eosin for two minutes. After air drying, sections were dehydrated in 95% and absolute alcohols, cleared in xylene, and coverslipped with Permount.

2.2. Cortical layer measurements

Linear measurements of cortical thickness from each tissue sample were obtained by using a stage micrometer and a camera lucida attachment on a Nikon Optiphot II microscope with a final magnification of 50×. Measurements were taken from flat sections cut orthogonal to the cortical surface, avoiding crowns and deep sulci. Gyral folds, where the cortical sheet can be distorted, were specifically avoided. Ten linear measurements of each cortical region were taken from each species studied, as multiple examples of the same species were often available. All linear measurements followed the radiation of the cortical columns.

Measurements of the specimens from the Yakovlev collection were taken from drawings made using a camera lucida attachment on an Olympus microscope at the Walter Reed Army Medical Center at a final magnification of 40×. Raw measurements were acquired by optically scanning the drawings (Fugitsu, Inc.) and saving the digital files at their maximum resolution. Unscaled raw measurements in millimeters were recorded by using on-screen linear measurements.

For the purposes of the present study, layer measurements followed the conventional six-layer system. In species where it was difficult to reliably distinguish between specific individual layers, such as layers II and III in the rodent [3,6,67–69], individual layers were combined into a composite measure (e.g., supragranular layer). In contrast, layer IV measures were often omitted when they could not be reliably visualized (e.g., in primary motor/agranular cortex).

2.3. Data analysis

To evaluate differences between orders for both the pyramidal and nonpyramidal layers, mixed model analysis of variance was employed. To control for overall size differences across species, absolute measurements of each layer were size-normalized to the entire cortical thickness. In order to evaluate the impact of other species-specific characteristics on the pattern of cortical layering, previously published data on brain weight, body weight, gestation period, average litter size, longevity, neurogenesis timespan, and encephalization quotient (EQ) were obtained from multiple sources [14–17,23,28,33,46,50,55,71,74,81]. Neurogenesis timespan data was only available for the mouse, rat, cat, ferret, rhesus macaque, and human species [14,15,17]. When published values differed, the most recent, complete, and comprehensive data source was given priority. To further reduce variance of related measures, brain weight and body weight for each individual species was always taken from the same primary source. These values were then used to directly calculate an EQ for each species [33] using the following formula: EQ = Measured Brain Weight / Expected Brain Weight, where Expected Brain Weight is equal to .12 × (Body Weight)$^{2/3}$. Independent contrasts were used to control for phylogenetic inertia between species [29,30], and the phylogeny required for this analysis was drawn from several sources [11,24,44,59,82].

3. Results

3.1. Pyramidal cell layers

Our primary analysis focused upon the degree to which total cortical thickness is devoted to intracortical connectivity as assessed by supragranular layer thickness and extracortical connectivity as assessed by infragranular layer thickness. Since only data from the primary sensory region was available for every species studied, we used this region for the main analysis. An analysis of variance (ANOVA) of the raw layer measurements examining mammalian order (primate, carnivore, and rodent) as well as layer (supragranular vs. infragranular layers) revealed a significant interaction between layer thickness and order (F = 34.75, df = 2, P < .001; see Fig. 2A). Additionally, there was a significant difference between layers (F = 9.13, df = 1, p = .005) but no difference between orders (F = 1.86, df = 2, p = .174). The difference between the absolute thickness of the supragranular layers ($\bar{x}$ = 624.4 μm) and the infragranular layers ($\bar{x}$ = 740.7 μm) is not surprising, while the lack of an overall difference in the absolute thickness between the orders examined is likely due to the unusually thick cortices of certain rodent species (e.g., porcupine and woodchuck) combined with the high variability within orders (see Fig. 3A). Of primary interest to the present study, however, is the magnitude of the interaction between layer thickness and mammalian order. Such a result demonstrates that, relative to infragranular measures, supragranular thicknesses are largest in primates ($\bar{x}$ = 867.9 μm) as compared to carnivores ($\bar{x}$ = 567.1 μm) and rodents ($\bar{x}$ = 438.2 μm), while infragranular thicknesses are largest in rodents (906.0 μm) as compared to primates (670.3 μm) and carnivores (645.9 μm; see Figs. 2A and 3A).

To control for size differences between species, the same analysis was conducted on values of the supragranular and infragranular layers that were normalized to total cortical thickness. As with the raw measures, proportional measures of supragranular and infragranular layer thickness interacted

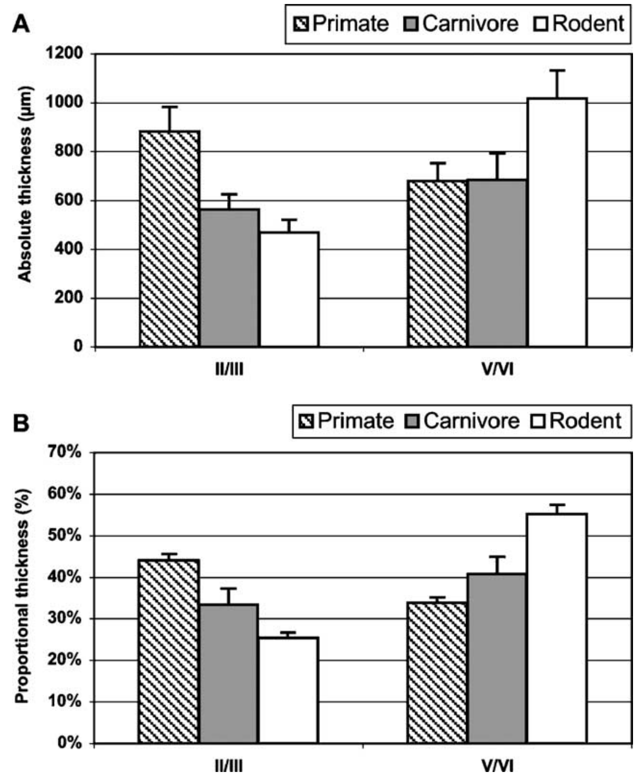

Fig. 2. Absolute (A) and proportional (B) measures of surpagranular (Layers II/III) and infragranular (Layers V/VI) thicknesses across the three orders. Supragranular layer thickness is largest in primates and smallest in rodents even after controlling for total cortical thickness.

with mammalian order (F = 55.38, df = 2, P < .001) and differed from each other (F = 17.16, df = 1, P < .001) and between the three orders (F = 4.27, df = 2, P = .024). On average, 44 percent of the total cortical thickness in primates is devoted to the supragranular layers, while in carnivores and rodents these layers occupy 35 and 26 percent of the total cortical thickness, respectively (see Fig. 2B). The infragranular layers show a complementary pattern occupying 34 percent of the total cortical thickness in primates, 39 percent in carnivores, and 54 percent in rodents. Unlike measures of total cortical thickness, the three orders examined show little overlap in the proportional size of the supragranular layers (see Fig. 3B), suggesting a divergence in their organization.

3.2. Nonpyramidal layers

Raw measures of the marginal (layer I; $\bar{x}$ = 188.79; range 164.79–202.57) and granular cell (layer IV; $\bar{x}$ = 206.67, range 167.22–234.37) layers were relatively uniform across the species examined (see Fig. 3A). An analysis of these layers showed a modest but nonsignificant effect of order (F = 3.12, df = 2, P = .059) and no interaction between order and layer (F = .791, df = 2, P = .463), as well as no difference between the two layers (F = 2.521, df = 1, P = .123; see Fig. 4A). Normalized measures of layer I ($\bar{x}$ = .107; range .098–.121) and layer IV ($\bar{x}$ = .119, range .100–.135)

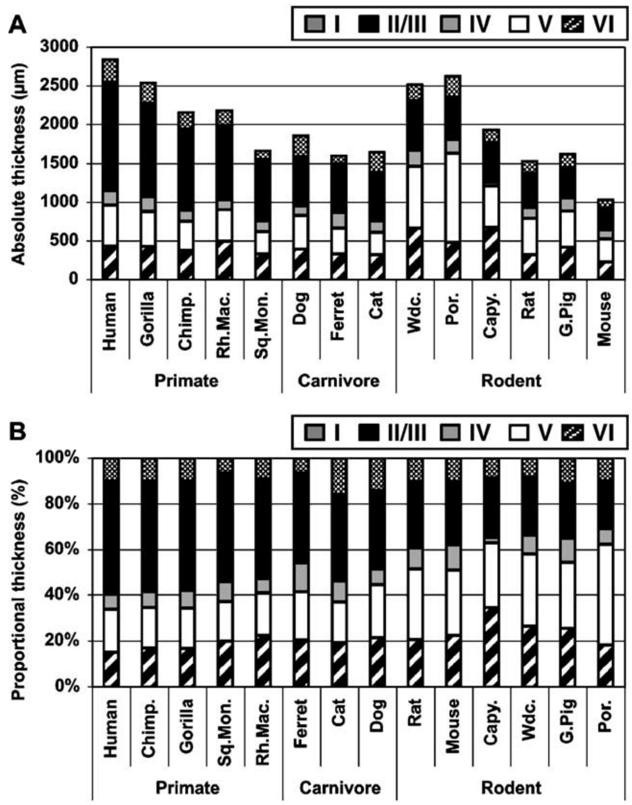

Fig. 3. Absolute (A) and proportional (B) measures of individual layer thicknesses in all 14 of the species examined. Despite variability in the thickness of cortex within the three mammalian orders represented, normalized measures of the laminar pattern show differences amongst the species in the proportion of cortex devoted to the supragranular layers with little overlap in this measure between the orders.

differed across the orders ($F = 4.269$, $df = 2$, $P = .024$) but did not differ substantially from each other ($F = 3.088$, $df = 1$, $P = .089$; see Fig. 4B). Additionally, there was no interaction between the layers and the orders ($F = .864$, $df = 2$, $P = .432$). Thus, modest differences across the orders were present in both the absolute and relative size of these layers, although each layer accounted for only ten to fourteen percent of total cortical thickness depending upon the species examined. Relative to the pyramidal cell layers, the contribution of these non pyramidal layers is more uniform across the orders (however see [47] for species differences in layer I, where a larger sampling of terrestial and aquatic mammals is considered).

3.3. Phylogenetic contrasts

Expansion in the relative thickness of the supragranular layers across orders could be a function of changes in absolute brain size, since these layers are responsible for corticocortical connections. Alternatively, supragranular layer expansion could also be a reflection of changes in the complexity of connections in the corticocortical layers that are independent of absolute brain size. As expected, raw measures of supragranular layer thickness were highly correlated with brain weight ($r^2 = .781$, $P < .0001$), encephalization quotient ($r^2 = .659$, $P = .0004$), gestation time ($r^2 = .652$, $P = .0005$), and neurogenesis timespan ($r^2 = .976$, $P = .0001$). Normalized values of supragranular layer thickness were also highly correlated with brain weight ($r^2 = .481$, $P = .006$), encephalization quotient ($r^2 = .486$, $P = .0056$), gestation time ($r^2 = .333$, $P = .03$), and neurogenesis timespan ($r^2 = .9418$, $P = .0013$).

Because phylogenetic inertia limits the amount that morphological measures can vary between species [29,30], independent contrasts were utilized to determine the associations between these variables that were not attributable to phylogenetic relatedness [60]. Raw measures of supragranular layer thickness were still correlated with measures of species size such as brain weight ($r^2 = .770$, $P < .001$; Fig. 5A) and encephalization quotient ($r^2 = .331$, $P = .050$; Fig. 5B), and were also correlated with developmental measures such as gestation time ($r^2 = .313$, $P = .059$; Fig. 5C) and neurogenesis timespan ($r^2 = .898$, $P = .004$; Fig. 5D). In contrast, normalized measures of supragranular layer thickness showed little relationship to brain weight ($r^2 = .089$, $P = .348$; Fig. 5E), encephalization quotient ($r^2 = .122$, $P = .226$; Fig. 5F), and gestation time ($r^2 = .021$, $P = .655$; Fig. 5G). Notably, neurogenesis time span and normalized supragranular layer thickness

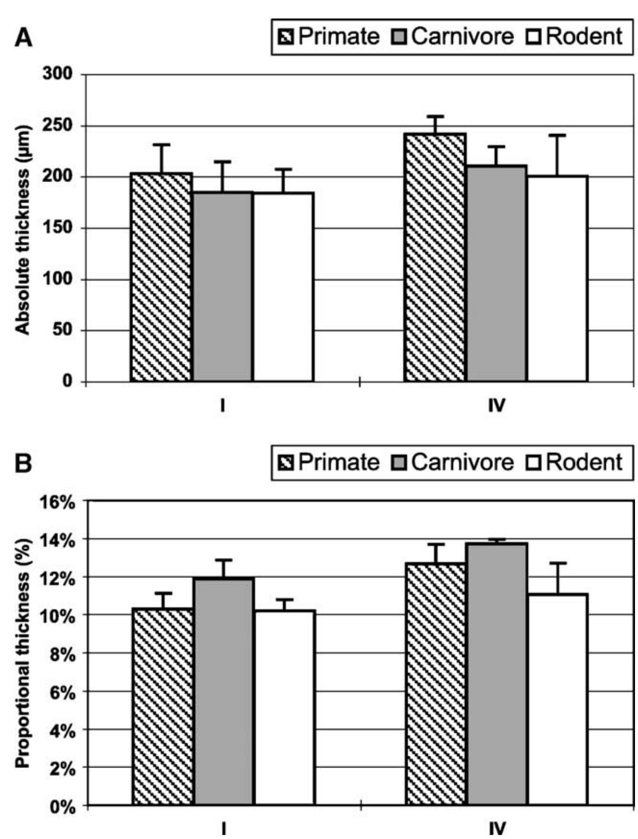

Fig. 4. Absolute (A) and proportional (B) measures of layer I and layer IV thicknesses across the three orders. Each layer occupies approximately 10 to 14 percent of the total cortical thickness with carnivores having the largest proportional values, but primates having the largest absolute values.

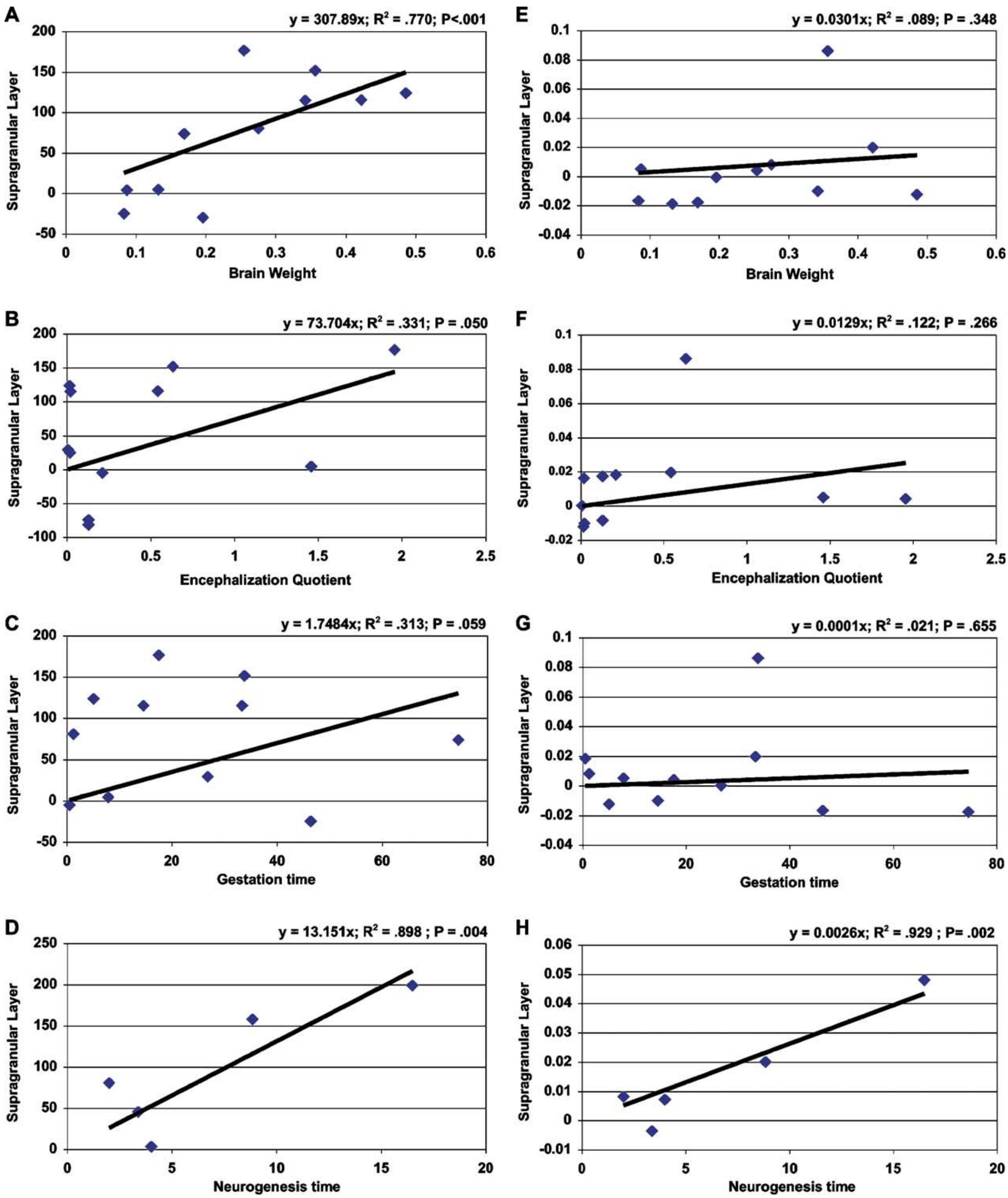

Fig. 5. Independent contrasts between absolute (left column) and proportional (right column) measures of supragranular layer thickness and measures of species brain weight (A, E), encephalization quotient (B, F), gestation time (C, G), and neurogenesis timespan (D, H). Note that proportional measures of supragranular layer thickness are only related to neurogenesis timespan, while absolute measures are correlated with each of these variables.

continued to be highly correlated (r^2 = .929, P = .002; Fig. 5H). Thus, after controlling for phylogentic relatedness, raw measures of layer thickness were always correlated with measures of overall size and development time, while the pattern of layering as assessed by relative layer thickness was only related to neurogenesis time span.

3.4. Regional measures

To explore whether these layering results were generalizable, motor, premotor, and somatosensory cortex were examined in a subset of cases that included every species except mouse. Raw measures of regional thickness differed

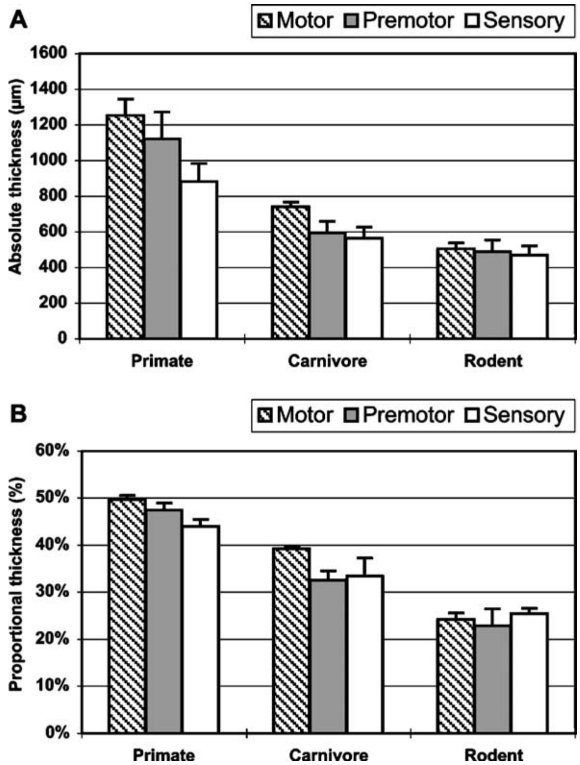

Fig. 6. Proportional measures of the supragranular (A) and infragranular (B) layers for motor, premotor and somatosensory cortex compared across the three orders. Each of these layers only varied modestly across the regions within an order.

from each other (F = 9.63, df = 2, P < .001) despite there being no difference in average cortical thickness across the three orders (F = 1.97, df = 2, P = .169). Motor cortex ($\bar{x}$ = 2151.26), on average, was thicker than either premotor ($\bar{x}$ = 1978.20) or sensory locations ($\bar{x}$ = 1861.10), a pattern that was the same in both primates and carnivores. In rodents, the premotor region was smaller than both the motor and sensory locations as was evident from a significant interaction between order and region (F = 4.981, df = 4, P = .003). These results may be flawed, however, since raw measures of cortical and layer thickness taken from histological sections will always overestimate true cortical thickness due to the inability to remain orthogonal to the layers in three dimensions.

Although similar across the three cortical regions sampled, the pattern of layering, as assessed by normalized thickness measures, did vary. As expected, layer IV was proportionally larger (.13) in sensory regions as compared to motor (.042) and premotor (.067) locations, while measures of the marginal zone were fairly consistent across locations (motor = .110, premotor = .130, sensory = .112). Consistent with alterations in the relative thickness of layer IV, the combined proportion of the pyramidal cell layers (supragranular + infragranular) was always greatest in motor regions (primates = .885, carnivores = .817, and rodents = .838) and always smallest in primary sensory regions (primates = .775, carnivores = .736, and rodents = .763). Finally, the proportion of cortex devoted to the supra- and infragranular layers was consistent across the cortical locations examined in the current study (see Fig. 6).

4. Discussion

The present study documents layering differences between orders of mammalian species, while layering patterns within orders are relatively uniform. Once normalized to total cortical thickness, the supragranular layers occupy 46 (±3; s.e.m.) percent of the primate cortex, 36 (±3) percent of the carnivore cortex, and 19 (±4) percent of the rodent cortex, with little overlap in values between the orders. One might expect morphometric variables of total brain size to be highly correlated with proportional layer thickness, since the supragranular layers are primarily responsible for intracortical connectivity. Although both the raw and proportional thickness of the supragranular cell layers are highly correlated with overall brain size, once phylogentic distance (inertia) is taken into account, supragranular layer thickness is most highly associated with ontogenetic variables such as neurogenesis time span and not measures of overall size.

The laminar patterns within the cerebral cortex and the organization of individual cortical columns are tightly associated. The organization of these columnar units, as well as their afferent and efferent connections, determines their packing density, their subsequent laminar pattern, and, ultimately, the total thickness of cortex [34]. Size normalization reveals that while species within an order may vary greatly in body size, brain weight, and cortical thickness (e.g., mouse vs. capybara), the proportion of their cortex devoted to the supragranular layers is similar and has little overlap with values found in species drawn from other orders. Additionally, the combined supra and infragranular (pyramidal) layers occupy a similar proportion of cortex within all of the species examined, although the ratio between these layers can change substantially. These findings, along with previous studies [4,5,20,21,34,47], underscore the notion that cortical layering and columnar organization is substantially altered between major phylogentic branches in mammalian evolution [57,58]. Rockel et al. also discussed differential laminar structure between species together with their cortical uniformity hypothesis (see Rockel et al. [72], Figs. 3–6). However, their data were not size-normalized, leaving the discussion open to a confound that originates from the positive relationship between brain size and cortical thickness [3,46].

The supragranular layers are comprised of layers II and III of the conventional six layer system [13,49]. They mature last during cortical development and establish a network of corticocortical projections [25,34]. Due to their connections with intra- and interhemispheric cortical locations, and the supragranular layers more generally, supragranular pyramidal cells have long been associated with the cognitive capabilities of individual species, especially primates. In older writings, the neurons of these layers are called "psychic" cells to indicate their importance in interconnecting diverse cortical regions and in potentially establishing functional networks of regions in a variety of cognitive tasks [51,69]. Differential laminar thickness in these layers may imply an unequal degree of connectivity [21,34] that could play a role in the cognitive and behavioral differences of various species [20,38] and contribute to regional specializations within a species [40,61].

The current study partially supports the proposal that expansion of the cortical layers characterized by interconnections with other cortical regions might be associated with changes in behavioral complexity and/or social intelligence [70]. This generalization is, however, moderated significantly by the stability of supragranular layer proportions within the orders examined in the present work. If such a gross measure had a simple relationship with cognitive capacity and/or behavioral complexity, one might expect to see greater variability in supragranular layer proportions between species within an order [70].

Cortical size varies substantially both between species and between orders. Many previous authors have noted that the thousand-fold horizontal expansion in the cortical sheet is the major feature of mammalian cortical evolution, and in many cases have downplayed the significance of changes in overall cortical thickness [31,46,51,54,61,73]. The notion that changes to cognitive capacity are primarily a function of an increase in the number of uniform cortical columns [75] does not give full merit to the impact that a change in the organization of these units could potentially have on cortical processing as a whole [43,80].

The horizontal expansion of the cortical sheet is likely determined early in development by altering the number of symmetrical divisions within the ventricular zone that will ultimately determine the size of the cortical progenitor cell population [13]. Additionally, the later-occurring period of asymmetric cell division may determine the number of units within a cortical column. Thus, changes to the number of cell cycles during early and late phases of neurogenesis may determine both the final number of cortical columns as well as the number of units within a cortical column.

During neurogenesis the relative, but not absolute, time span allotted to the development of each layer appears to be uniform in the small number of species that have been examined [13]. This suggests that differential expansion of cortical layers across orders may be related to cell size and neuropil fraction changes rather than to changes in the number of cells within an individual vertical column as it traverses a specific layer. However, cell cycle rates contributing to individual layers of the cortex are not uniform over time. Cell cycles in the mouse slow as the superficial layer neruoblasts are being produced, while cell cycles slow initially in the *maccaca mulatta*, but then speed up during the generation of the most superficial layers, providing a potential mechanism for adding additional neuronal units and circuitry to these layers [36,48,80]. In agreement with such a model, neurogenesis time span in the present study correlates highly with both raw and normalized supragranular layer thickness, even when one employs the method of independent contrasts in an attempt to control for phylogenetic inertia in closely related species [29,30,60]. This finding is notable, since measures such as brain weight, encephalization quotient, and gestation time, which also correlate highly with supragranular layer thickness in standard analyses, do not remain correlated once phylogeny and total cortical thickness are controlled (see Fig. 5). It is therefore possible that changes to the absolute time of neurogenesis and, presumably, the number of cell cycles that occur during neurogenesis might underlie the pattern of lamination in a species and the expansion or contraction of the supragranular layers.

This relationship between neurogenesis and subsequent lamination patterns has been proposed in a number of contexts, and it has been noted that modifications in the timing of neurodevelopment events (i.e., neugenesis onset, neurogenesis offset, and cell cycle speeds) could produce relatively dramatic changes to cortical structure [17,23,62,66]. The importance of timing changes during development as a major mediator of biological change was recognized even in the early theories of biodiversity (see [27] for a historical review). In the context of cortical neurogenesis, early changes would primarily affect the expansion of the cortical sheet by altering the number of progenitors and columnar units within the cortex, while late changes would impact the internal organization of the individual columnar units and their subsequent aggregation within the developing cortical plate [13,66]. The present data shows relative stability in the lamination patterns within the three mammalian orders examined and, therefore, suggests that there is substantial constraint on the developmental processes that determine these patterns. This supports previous suggestions that across primate species development is allometric, and thus constrained [14].

A full understanding of differential mammalian cortical expansion in the vertical domain [72] depends upon a better understanding of the three-dimensional organization of adult cortex in a wide variety of species in terms of the number of cells in a column [47] and the proportion of cells within a column occupying an individual layer. In addition, one would also wish to understand a number of neurodevelopmental metrics, including total neurogenesis time, changes in the terminal mitosis fraction of proliferative cells in the ventricular zone [13], rates of cell division as they change over total neurogenesis time [37,64], and rates of maturation. Although the present study attempted to include species of varying size within three orders of eutherian mammals and several cortical locations, additional metrics from other cortical regions, especially association and limbic cortex [76], would be desirable. This may be especially important in primate species where the timing of neurogenesis may vary systematically across cortical locations in a manner that is related to areal identity and subsequent cortical organization ([65]; Fig. 2). In addition, the inclusion of a broader variety of mammals from unique ecologies [40] (e.g., aquatic mammals) can only help to clarify our understanding of cortical modification throughout evolutionary history and how such organization might be adapted to both unique ecological conditions and the functional demands of an individual cortical location [32].

In addition to a limited sample of mammals and a limited investigation of cortical regions, the findings of the current study are tempered by a number of methodological difficulties. Since several exotic species were included in this work, we were unable to control for agonal state, specific tissue fixation protocols, and, in cases where tissue had already been processed, the exact parameters of the Nissl stain employed. Our focus on proportional measures diminishes but does not eliminate concerns of differential shrinkage across species. In addition, we did not always have access to the entire brain specimen, and although we were able to obtain samples of primary somatosensory cortex from each species, in several cases samples of motor and premotor regions were not readily available. The precise location of these cortical regions was based upon available cortical maps and stereotaxic atlases; however, the equivalence of cortical locations across species can never be assumed [40]. Despite this, large differences in the proportional organization of the supra- and infragranular layers were not present across regions within the species utilized in the current study. Finally, the demarcation of layer boundaries was made by an experienced neuroanatomist (JH), but others have employed computer algorithms to determine the boundaries of individual cortical layers [8]. Given the diversity of the species utilized in the present study, an approach that is sensitive to multiple image analysis parameters would not be universally applicable in all cases.

5. Conclusions

The current study demonstrates that the proportion of cortex occupied by the supragranular layers varies among mammalian orders and is largest in primate species, smaller in carnivores, and smallest in rodents. The relative stability of the proportion of cortex occupied by these layers *within* orders of mammals indicates that such divergence in the overall plan of cortical layering may have occurred early in the phylogeny of each order. Therefore, although layering patterns are relatively constrained from change over time, they are not immutable to change [20,25,26,36,51,57,63]. Specifically, subcomponents of cortical layering, especially those participating in cortico-cortical connections, may vary substantially amongst orders of mammals.

Acknowledgments

The authors would like to thank Archie Fobbs, curator of the mammalian brain collection at the National Museum of Health and Medicine in Washington, DC; Cara Schneider, formerly at the University of California, Davis; Dr. Stephan Bodnarenko of Smith College in Amherst, MA; and Dr. Joseph Marcus of Boston Medical School for their invaluable assistance in assembling the mammalian species utilized in this study. We would also like to thank Dr. Joseph Marcus for his expert advice and consultation during both the early and late phases of this project.

References

[1] D.L. Adams, J.C. Horton, Capricious expression of cortical columns in the primate brain, Nat. Neurosci. 6 (2003) 113–114.
[2] O.A. Adrianov, T.A. Mering, Atlas of the Canine Brain, Edwards Bros., Ann Arbor, MI, 1964.
[3] P. Bailey, G.V. Bonin, The Isocortex of Man, vol. 6, University of Illinois Press, Urbana, 1951.
[4] P. Barone, H. Kennedy, Non-uniformity of neocortex: Areal heterogeneity of NADPH-diaphorase reactive neurons in adult macaque monkeys, Cereb. Cortex 10 (2000) 160–174.
[5] C. Beaulieu, Numerical data on neocrotical neurons in adult rat with special reference to the GABA population, Brain Research 609 (1993) 284–292.
[6] H. Braak, Architectonics of the Human Telencephalic Cortex, vol. 4, Springer-Verlag, New York, 1980.
[7] D.P. Buxhoeveden, M.F. Casanova, The minicolumn hypothesis in neuroscience, Brain 125 (2002) 935–951.
[8] D.P. Buxhoeveden, A.E. Switala, E. Roy, M.F. Casanova, Quantitative analysis of cell columns in the cerebral cortex, J. Neurosci. Methods 97 (2000) 7–17.
[9] D.P. Buxhoeveden, A.E. Switala, E. Roy, M. Litaker, M.F. Casanova, Morphological differences between minicolumns in human and non-human primate cortex, Am. J. Phys. Anthropol. 115 (2001) 361–371.
[10] G.B. Campos, W.I. Welker, Comparisons between brains of a large and a small hystricomorph rodent: Capybara, Hydorchoerus and guinea pig, Cavia; neocortical projection regions and measurements of brain subdivisions, Brain Behav. Evol. 13 (1976) 243–266.
[11] M.D. Carleton, Introduction to rodents, in: S. Anderson, J.K. Jones (Eds.), Orders and Families of Recent Mammals of the World, John Wiley and Sons, New York, 1984, pp. 255–265.
[12] M.F. Casanova, D. Buxhoeveden, G.S. Sohal, Brain development and evolution, in: M. Ernst, J.M. Rumsey (Eds.), Functional Neuroimaging in Child Psychiatry, Cambridge University Press, Cambridge (UK), 2000, pp. 113–136.
[13] V.S.J. Caviness, T. Takahashi, R.S. Nowakowski, Numbers, time and neocortical neurogenesis: a general developmental and evolutionary model, Trends Neurosci. 18 (1995) 379–383.
[14] B. Clancy, R.B. Darlington, B.L. Finlay, The course of human events: predicting the timing of primate neural development, Dev. Sci. (2000).
[15] B. Clancy, R.B. Darlington, B.L. Finlay, Translating developmental time across mammalian species, Neuroscience 105 (2001) 7–17.
[16] G.B. Corbet, J.E. Hill, A World List of Mammalian Species, Cornell University Press, Ithaca, N.Y, 1980.
[17] R.B. Darlington, S.A. Dunlop, B.L. Finlay, Neural development in metatherian and eutherian mammals: Variation and constraint, J. Comp. Neurol. 411 (1999) 359–368.
[18] M.R. DeLucchi, B.J. Dennis, W.R. Adey, A Stereotaxic Atlas of the Chimpanzee Brain, University of Califronia Press, Berkeley, CA, 1965.

[19] G.S. Doetsch, A.L. Towe, The pyramidal system of the woodchuck, Brain Behav. Evol. 19 (1981) 37–55.
[20] G.N. Elston, Cortex, cognition and the cell: New insights into the pyramidal neuron and prefrontal function, Cereb. Cortex 13 (2003) 1124–1138.
[21] G.N. Elston, M.G.P. Rosa, Pyramidal cells, patches and cortical columns: A comparative study of infragranular neurons in TEO, TE, and the superior temporal polysensory area of the macaque monkey, J. Neurosci. 20 (2000) RC117.
[22] R. Emmers, K. Akert, A Stereotaxic Atlas of the Brain of the Squirrel Monkey, University of Wisconsin Press, Madison, WI, 1963.
[23] B.L. Finlay, R.B. Darlington, Linked regularities in the development and evolution of mammalian brains, Science 268 (1995) 1578–1584.
[24] J.J. Flynn, M.A. Nedbal, Phylogeny of the carnivora: congruence vs. incompatibility among multiple data sets, Mol. Phylogenet. Evol. 9 (1998) 414–426.
[25] J.M. Fuster, Neurobiology of cortical networks, Cortex and Mind, Oxford University Press, New York, 2003, pp. 17–53.
[26] G.J. Goodhill, M.A. Carreira-Perpinan, Cortical columns, Encyclopedia of Cognitive Science, Macmillan Publishers Ltd, 2002.
[27] S.J. Gould, Ontogeny and Phylogeny, The Belknap Press of Harvard University Press, Cambridge, MA, 1977.
[28] Grzmak's Encyclopedia of Mammals, McGraw Hill, 1990.
[29] P.H. Harvey, M.D. Pagel, The Comparative Method in Evolutionary Biology, Oxford University Press, Oxford, 1991.
[30] P.H. Harvey, A. Purvis, Comparative methods for explaining adaptations, Nature 351 (1991) 619–624.
[31] M.A. Hofman, On the evolution and geometry of the brain in mammals, Prog. Neurobiol. 32 (1989) 137–158.
[32] J.J. Hutsler, R.A.W. Galuske, Hemispheric asymmetries in cerebral cortical networks, Trends Neurosci. 26 (2003) 429–435.
[33] H.J. Jerison, Evolution of the Brain and Intelligence, Academic Press, New York, 1973.
[34] E.G. Jones, Anatomy of cerebral cortex: Columnar input-output organization, in: F.O. Schmitt, F.G. Worden, G. Adelman, S.G. Dennis (Eds.), The Organization of the Cerebral Cortex, The MIT Press, Cambridge (MA), 1981, pp. 199–235.
[35] E.G. Jones, Microcolumns in the cerebral cortex, Proc. Natl. Acad. Sci. U. S. A. 97 (2000) 5019–5021.
[36] E.G. Jones, A. Peters, Comparative structure and evolution of cerebral cortex: Part I, in: E.G. Jones, A. Peters (Eds.), Cerebral Cortex, vol. 8A, Plenum Press, New York, 1984.
[37] D.R. Kornack, P. Rakic, Changes in cell-cycle kinetics during the development and evolution of primate neocortex, Proc. Natl. Acad. Sci. U. S. A. 95 (1998) 1242–1246.
[38] K. Kowalski, Mammals = Ssaki: an Outline of Theriology, U.S. Department of Commerce: National Technical Information Service, Springfield, VA, 1976.
[39] H.J. Kretschmann, G. Kammradt, E.C. Cowart, A. Hopf, I. Krauthausen, H.W. Lange, B. Sauer, The Yakovlev Collection: a unique resource for brain research and the basis for a multinational data bank, J. Hirnforsch. 23 (1982) 647–656.
[40] L. Krubitzer, The organization of neocortex in mammals: are species differences really so different? Trends Neurosci. 18 (1995) 408–417.
[41] L. Kruger, Photographic Atlas of the Rat Brain: The Cell and Fiber Architecture Illustrated in Three Planes with Stereotaxic Coordinates, Cambridge University Press, Cambridge, UK, 1995.
[42] B.I. Lockard, The forebrain of the ferret, Lab. Anim. Sci. 35 (1985) 216–228.
[43] R. Lorente de No, The cerebral cortex: Architecture, intracortical connections, motor projections, in: J.F. Fulton (Ed.), Physiology of the Nervous System, Oxford University Press, London, 1938, pp. 291–325.
[44] W.P. Luckett, J.-L. Hartenberger, Monophyly or polyphyly of the order rodentia: Possible conflict between morphological and molecular interpretations, J. Mammal Evol. 1 (1993) 127–147.
[45] T.J. Luparello, Stereotaxic Atlas of the Forebrain of the Guinea Pig, Williams and Willkins, Inc., Baltimore, MD, 1967.
[46] E.M. Macphail, Brain and Intelligence in Vertebrates, Oxford University Press, New York, 1982.
[47] J.A. Marcus, Radial Neuron Number and Mammalian Brain Evolution: Reassessing the Neocortical Uniformity Hypothesis, Doctoral Dissertation, Department of Anthropology, Harvard University, Boston, MA, 2003.
[48] M. Marin-Padilla, Ontogenesis of the pyramidal cell of the mammalian neocortex and developmental cytoarchitectonics: a unifying theory, J. Comp. Neurol. 321 (1992) 223–240.
[49] M. Marin-Padilla, Cajal-Retzius cells and the development of the neocortex, Trends Neurosci. 21 (1998) 64–71.
[50] D. Morris, The Mammals: a Guide to the Living Species, Harper and Row Publishers, 1965.
[51] V.B. Mountcastle, Modality and topographic properties of single neurons of cat's somatic sensory cortex, J. Neurophysiol. 20 (1957) 408–434.
[52] V.B. Mountcastle, An organizing principle for cerebral function: the unit modula and the distributed system, in: G.M. Edelman, V.B. Mountcaslte (Eds.), The Mindful Brain: Cortical Organization and the Group-Selective Theory of Higher Brain Function, MIT press, Cambridge(MA), 1978, pp. 7–51.
[53] V.B. Mountcastle, The columnar organization of the neocortex, Brain 120 (1997) 701–722.
[54] V.B. Mountcastle, Introduction: computation in cortical columns, Cerebral Cortex 13 (2003) 2–4.
[55] R.M. Nowak, Walker's Mammals of the World, The Johns' Hopkins University Press, 1991.
[56] G. Paxinos, The Rhesus Monkey Brain in Steretoxic Coordinates, Academic Press, San Diego, CA, 2000.
[57] T. Preuss, Preface: from basic uniformity to diversity in cortical organization, Brain Behav. Evol. 55 (2000) 283–286.
[58] T. Preuss, Taking the measure of diversity: Comparative alternatives to the model-animal paradigm in cortical neuroscience, Brain Behav. Evol. 55 (2000) 287–299.
[59] A. Purvis, A composite estimate of primate phylogeny., Philos. Trans. R. Soc. Lond. B. 348 (1995) 405–421.
[60] A. Purvis, A. Rambaut, Comparative analysis by independent contrasts (CAIC): an Apple Macintosh application for analysing comparative data, CABIOS 11 (1995) 247–251.
[61] P. Rakic, Developmental events leading to laminar and areal organization of the neocortex, in: F.O. Schmitt, F.G. Worden, G. Adelman, S.G. Dennis (Eds.), The Organization of the Cerebral Cortex, The MIT Press, Cambridge (MA), 1981, pp. 7–28.
[62] P. Rakic, Specification of cerebral cortical areas, Science 241 (1988) 170–176.
[63] P. Rakic, A small step for the cell, a giant leap for mankind: a hypothesis of neocortical expansion during evolution, Trends Neurosci. 18 (1995) 383–388.
[64] P. Rakic, Radial unit hypothesis of neocortical expansion, Novartis Found. Symp. 228 (2000) 30–42.
[65] P. Rakic, Neurogenesis in adult primate neocortex: an evaluation of the evidence, Nature Rev. Neurosci. 3 (2002) 65–71.
[66] P. Rakic, Developmental and evolutionary adaptations of cortical radial glia, Cereb Cortex 13 (2003) 541–549.
[67] S. Ramon y Cajal, Cajal on the Cerebral Cortex, Oxford University Press, New York, 1988.
[68] S. Ramon y Cajal, The Cerebral Cortex, New Ideas on the Structure of the Nervous System in Man and Vertebrates, The MIT Press, Cambridge (MA), 1990, pp. 35–72.
[69] S. Ramon y Cajal, Histology of the Nervous System, vol. 2, Oxford University Press, New York, 1995, (429–492 pp.).
[70] S.M. Reader, K.N. Laland, Social intelligence, innovation, and enhanced brain size in primates, Proc. Natl. Acad. Sci. U. S. A. 99 (2002) 4436–4441.
[71] S.R. Robinson, B. Dreher, The visual pathways of the eutherian

mammlas and marsupials develop accroding to a common timetable, Brain Behav. Evol. 36 (1990) 177–195.
[72] A.J. Rockel, R.W. Hiorons, T.P.S. Poweel, The basic uniformity in structure of the neocortex, Brain 103 (1980) 221–224.
[73] J.L.R. Rubenstein, P. Rakic, Genetic control of cortical development, Cereb. Cortex 9 (1999) 521–523.
[74] I.S. Russell, Brain size and intelligence: a comparative perspective, in: D.A. Oakley, H.C. Plotkin (Eds.), Brain Behaviour and Evolution, Methuen, London, 1979, pp. 126–153.
[75] T. Sawaguchi, K. Kubota, A hypothesis on the primate neocortex evolution: column-multiplication hypothesis, Int. J. Neurosci. 30 (1986) 57–64.
[76] K. Semendeferi, E. Armstrong, A. Schleicher, K. Zilles, G. Van Hoesen, Limbic frontal cortex in hominoids: a comparative study of Area 13, Amer. J. Phys. Anthropol. 106 (1998) 129–155.
[77] R.L. Sidman, J.B. Angevine, E.T. Pierce, Atlas of the Mouse Brain and Spinal Cord, Harvard University Press, Cambridge, MA, 1971.
[78] T.S. Skoglund, R. Pascher, C.-H. Berthold, Heterogeneity in the columnar number of neurons in different neocortical areas in the rat, Neurosci Lett. 208 (1996) 97–100.
[79] R.S. Snider, W.T. Niemer, A Stereotaxic Atlas of the Cat Brain, University of Chicago Press, Chicago, IL, 1961.
[80] H. Super, H.B. Uylings, The early differentiation of the neocortex: a hypothesis on neocortical evolution, Cereb. Cortex 11 (2001) 1101–1109.
[81] The Encyclopedia of Mammals, Facts on File Publications, New York, 2001.
[82] T.A. Vaughan, N.J. Ryan, N.J. Czaplewski, Mammalogy, Saunders College Publishing, Philadelphia, 2000.